

PROBLEM SET 4

ECON301—ECONOMETRICS I

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Instructions: Due on 13/04/2021; scan or type your work and submit it through the link in Moodle. You are encouraged to work in groups but each student is required to submit her or his own work. If you worked with a group, please indicate the names of each member in your submission.

Note: in the lecture notes and videos, I usually denote vectors by underlining them, for example $\underline{x}$, and I denote matrices in capital letters. When typesetting it, I use **bold notation and upright letters** to denote vectors and matrices, i.e., $\mathbf{x}$ and $\mathbf{M}$ for example.

1. Consider the normal equations that determine the LSE: $(\mathbf{X}^\top \mathbf{X})\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$. Let $\mathbf{S}_{xx}$ be the $k \times k$ in which the (i, j) element is $S_{x_i, x_j} = \sum_{\ell=1}^n (X_{\ell i} - \bar{X}_i)(X_{\ell j} - \bar{X}_j)$, $\mathbf{S}_{xy}$ be the k dimensional vector with elements given by $S_{x_i, y} = \sum_{\ell=1}^n (X_{\ell i} - \bar{X}_i)(Y_{\ell} - \bar{Y})$, and $\hat{\boldsymbol{\beta}}_P = (\hat{\beta}_1, \dots, \hat{\beta}_k)^\top$ be the LSE estimator of the vector of slopes of the model. Show that $\mathbf{S}_{xx}\hat{\boldsymbol{\beta}}_P = \mathbf{S}_{xy}$.
2. Let $\mathbf{X}$ be a regression matrix such that $\mathbf{1} \in \mathcal{L}(\mathbf{X})$. Show that $\bar{Y} = \widehat{\bar{Y}}$ and $\widehat{\bar{\epsilon}} = 0$. *Hint: you can assume without loss of generality, that the vector of ones, $\mathbf{1}$, is part of the regression matrix $\mathbf{X}$.*
3. Let $(X_i, Y_i), i = 1, 2, \dots$ be independent observations that fulfill assumptions 1M and 2M. Let $\hat{\boldsymbol{\beta}}$ be the LSE estimator in a simple linear regression. Find and justify the limit in probability of $\frac{1}{n} \sum_{i=1}^n Y_i^2$, $\frac{1}{n} \sum_{i=1}^n \hat{\beta}_0 Y_i$, and $\frac{1}{n} \sum_{i=1}^n \hat{\beta}_1 Y_i X_i$.
4. Finish the proof of Theorem 5.2.
5. The purpose of this exercise is to show that we always have

$$\text{SSE}(X_1, \dots, X_k, X_{k+1}) \leq \text{SSE}(X_1, \dots, X_k).$$

Consider

$$\text{SS}(c_0, c_1, \dots, c_k, c_{k+1}) = \frac{1}{n} \sum_{i=1}^n (y_i - c_0 - c_1 x_{i,1} - c_k x_{i,k} - c_{k+1} x_{i,k+1})^2,$$

and provide an argument using the idea of constrained optimization.